Amarthya Jonnalagadda

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Summary

Final-year CSE (AI & ML) student who builds, fine-tunes and evaluates AI systems: a live multi-agent LLM platform, a Telugu voice assistant with a fine-tuned vision model (paper under review), and benchmarks of LLM fine-tuning and test quality. Full-stack internship (Java, Spring Boot, React). Seeking **AI/ML Engineer** or **Software Engineer** roles.

Education

VIT-AP University, B.Tech in Computer Science Engineering (AI & ML) 2023 – 2027
- **CGPA: 8.55/10** | **Coursework:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, ML

Dr. KKR's Happy Valley School – Class XII: 88% 2023
KKR Gowtham School, Nellore – Class X: 100% 2021

Experience

Full Stack Web Developer Intern, GRAAS (Growth as a Service), Coimbatore June 2025 – July 2025
- Developed 2 full-stack web applications end-to-end in **Java, Spring Boot and React** for internal and client use
- Designed and implemented **10+ RESTful API endpoints** and integrated them with responsive React interfaces
- Debugged and resolved backend and frontend defects, reducing reported issues during QA cycles

Projects

Pathlight – Autonomous Career Intelligence Platform (Agentic AI, Solo) [Live] [GitHub Repo]
- Deployed an agentic AI platform that sources jobs from boards and placement emails, checks eligibility, finds skill gaps, tailors resumes, and applies **only after the user approves**
- Orchestrated **5 LangGraph agents** with **deterministic rules before any LLM call**, a multi-provider LLM fallback chain and **4 MCP servers** for tool calling (Gmail, GitHub, filesystem, reminders)
- Wrote an **LLM evaluation harness** on hand-labeled benchmarks: discovery agent 100% field accuracy and 0.94 skills F1; eligibility agent 87.5% accuracy with **0 false positives**; used it to retune RAG retrieval thresholds
- Added scam detection (AICTE red flags), placement-policy eligibility rules and deduplicated deadline reminders
- **320+ automated tests** (pytest, Vitest) with fake-LLM injection and CI; a guard rejects invented skills
- *Tech: Python, FastAPI, LangGraph, MCP, Gemini API, RAG (ChromaDB), MongoDB, Next.js, TypeScript, Docker*

Krishi-Agent – Telugu Voice Farm Advisory (Team, Speech, Vision) [GitHub Repo]
- Voice assistant that answers farmers in spoken Telugu: speech recognition (ASR), a rules-first intent router, disease, weather and mandi-price agents, and a rule-based conflict resolver (paper under review)
- **Fine-tuned a DINOv2 ViT-L/14** crop-disease classifier in PyTorch: **23.5% to 72.9%** accuracy on PlantDoc field photos and **91.1%** on 2,640 unseen potato photos, with calibrated confidence
- Rules-first intent router hit **95.8%** vs 82.3% for the LLM alone (McNemar $p = 0.0002$); Telugu ASR at **11.7% WER**
- *Tech: PyTorch, DINOv2, FastAPI, Ollama (Qwen2.5-7B), AI4Bharat ASR/TTS, data.gov.in API*

PEFT Benchmark – LoRA vs DoRA vs QLoRA Fine-Tuning for Telugu [GitHub Repo]
- Fine-tuned Qwen3-0.6B on Aya Telugu with LoRA, DoRA and QLoRA (PyTorch, Hugging Face PEFT, bitsandbytes) on a T4 GPU: DoRA matched LoRA's test loss at **1.8x the training time**; 4-bit QLoRA saved no memory

Catchscore – Mutation Testing for AI-Written Code [GitHub Repo]
- Python CLI that injects bugs (AST mutations) into the lines a PR changed and runs its tests; `--suggest` has Claude write tests for the gaps, kept only if they pass and catch a bug
- Benchmarked **568 merged PRs** on GitHub Actions: tests let a median **17%** of injected bugs through, with no agent-vs-human gap once PR size is controlled

Quotation Management Web App (Full Stack, Real Client) [GitHub Repo]
- Digitized product and quotation management for a family-owned home-theatre business with **Spring Boot, React and MongoDB**, replacing a manual Excel process with automated PDF quotations; deployed on **Microsoft Azure**

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL
AI / LLM: LLMs, LangGraph, Multi-Agent Systems, MCP, RAG, Tool Calling, LLM Evaluation, Prompt Engineering
Machine Learning: PyTorch, Hugging Face Transformers, Fine-Tuning (LoRA, QLoRA, PEFT), Computer Vision (DINOv2, ViT), Model Calibration, Statistical Testing (McNemar), Speech (ASR/TTS)
Backend & Frontend: FastAPI, Spring Boot, REST APIs, JWT Authentication, React, Next.js
Data, Cloud & DevOps: MongoDB, SQL, Docker, Azure, AWS, Vercel, Git, GitHub Actions (CI/CD), pytest, Vitest

Certifications

- AWS Certified Cloud Practitioner [View Badge] | Scrimba Learn React Certification [View Certificate]